

FILTER DESIGN TEST CASE

FBAR (Film Bulk Acoustic Resonator) - Band 8 (900 MHz) for UWB (Ultra-Wideband) Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a FBAR (Film Bulk Acoustic Resonator) filter targeting Band 8 (900 MHz) for UWB (Ultra-Wideband) Filter. The filter is designed on ScAlN on Si substrate with a target center frequency of 3982.8 MHz and bandwidth of 314.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the UWB (Ultra-Wideband) Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: FBAR (Film Bulk Acoustic Resonator)
- Frequency Band: Band 8 (900 MHz)
- Application: UWB (Ultra-Wideband) Filter
- Substrate: ScAlN on Si
- Package: Flip-Chip BGA
- Design Tool: COMSOL Multiphysics 6.2
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the FBAR (Film Bulk Acoustic Resonator) design targeting Band 8 (900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3982.8 MHz
Bandwidth (3 dB):	314.8 MHz
Insertion Loss Target:	≤ 1.2 dB
Return Loss Target:	≥ 16.0 dB
Out-of-Band Rejection:	≥ 33.0 dB
Isolation (Duplexer):	≥ 31.0 dB
Q Factor:	10419
Temperature Coefficient:	-21.4 ppm/°C
Die Size:	0.8 x 1.1 mm
Wafer Size:	6-inch
Number of Resonators:	9
Electrode Material:	W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open COMSOL Multiphysics 6.2 and load the FBAR (Film Bulk Acoustic Resonator) template project.
2. Define the substrate stack: ScAlN on Si with specified layer thicknesses.
3. Set design targets: center freq 3982.8 MHz, BW 314.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, ScAlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 11948 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.2 dB
- Return loss in passband shall be >= 16.0 dB
- Out-of-band rejection at +/- 472 MHz offset >= 33.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 10.7 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 97%

6. TEST RESULTS

Measurement Date: 2024-02-07
Devices Tested: 91
Insertion Loss (meas): 0.9 dB
Return Loss (meas): 18.0 dB
Rejection (meas): 36.0 dB
Isolation (meas): 35.6 dB
Q Factor (meas): 10419
Group Delay Variation: 6.9 ns
Power Handling: +29 dBm
Die Yield: 82.0%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The FBAR (Film Bulk Acoustic Resonator) design demonstrated excellent performance for Band 8 (900 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Group delay flatness improved compared to previous revision.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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